

Diagnostics meets Genealogy: Evaluation of Sequence Variations in the Sheep Poodle Genome

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Introduction

Cryptorchidism is a common disorder in testicle development in male dogs, which affects up to 10% of them. In previous work, genes potentially involved in cryptorchidism were amplified and sequenced^{1, 2}. The overall goal of this study is to detect genetic variations in the sheep poodle genome in order to infer its relationships with other dogs. In particular, we focus on the genetic variation of candidate genes, in order to find a potential genetic origin of cryptorchidism.

Objectives

- Test whether candidate gene amplicon reads can be used to infer the pedigree of analysed individuals.
- Try to place the sheep poodle in a tree of dog breeds.

WP1: Genealogy

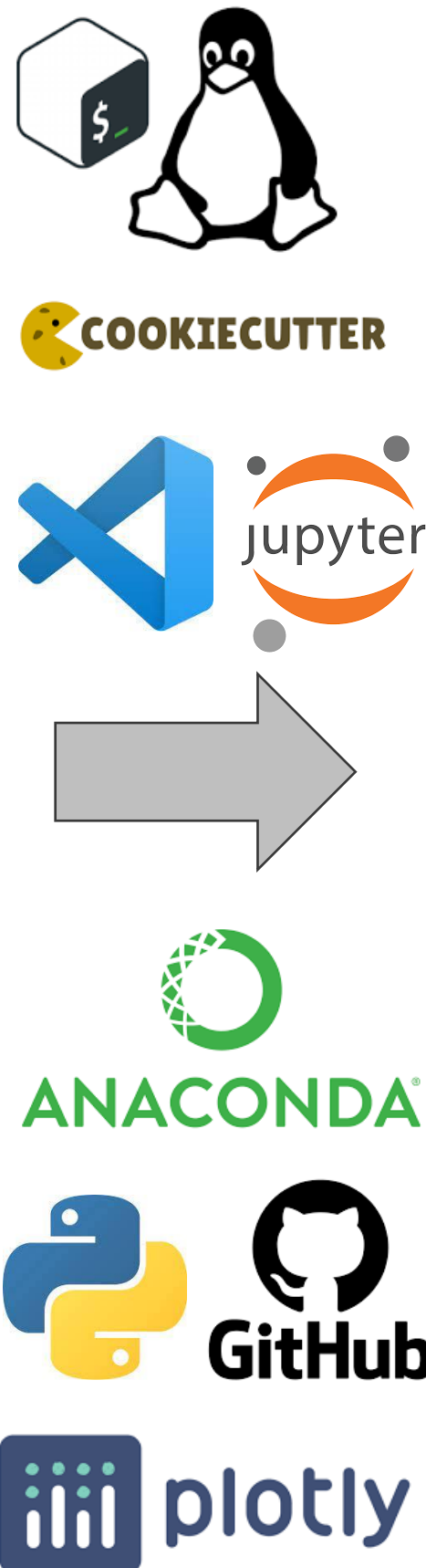
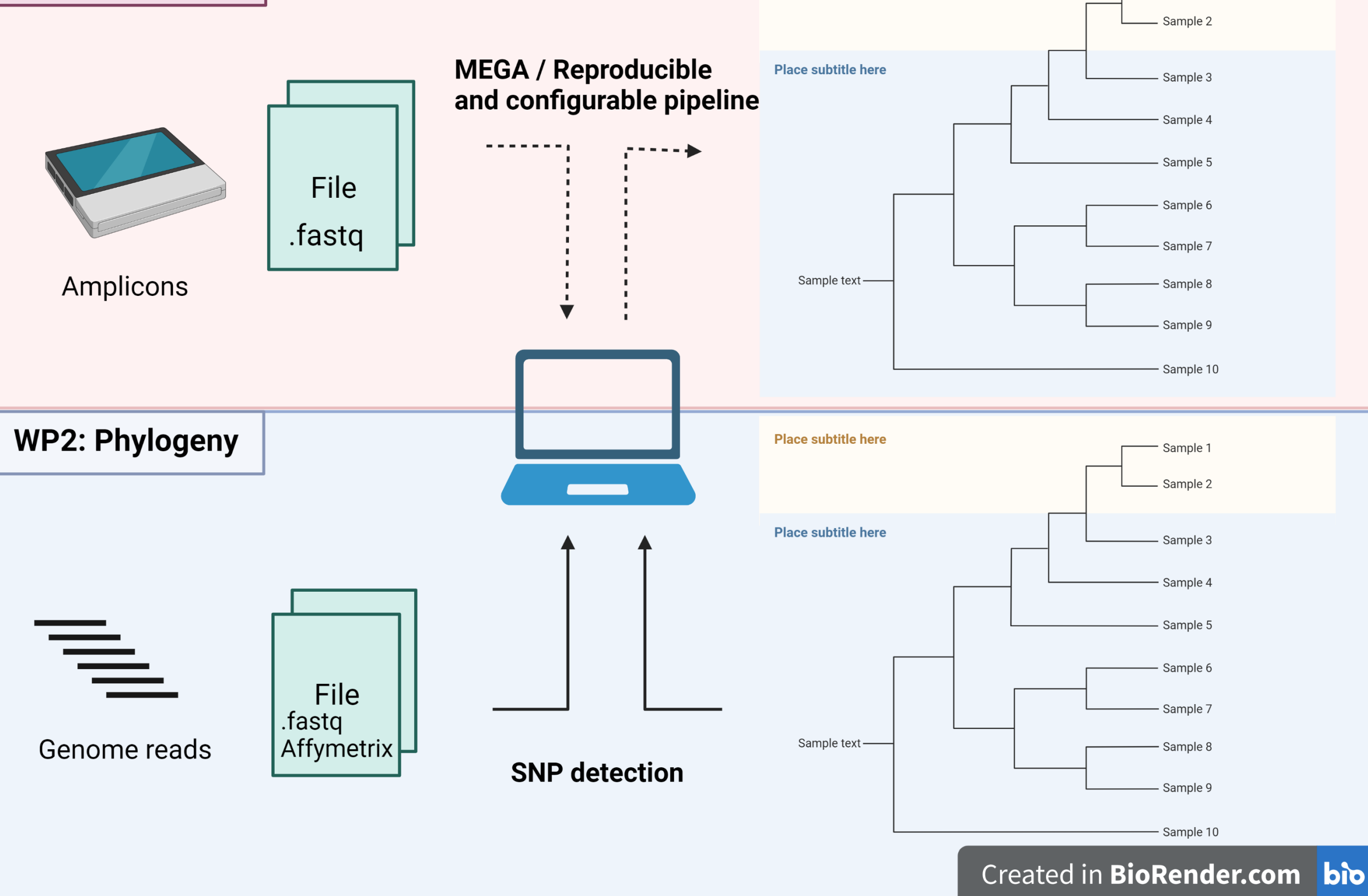
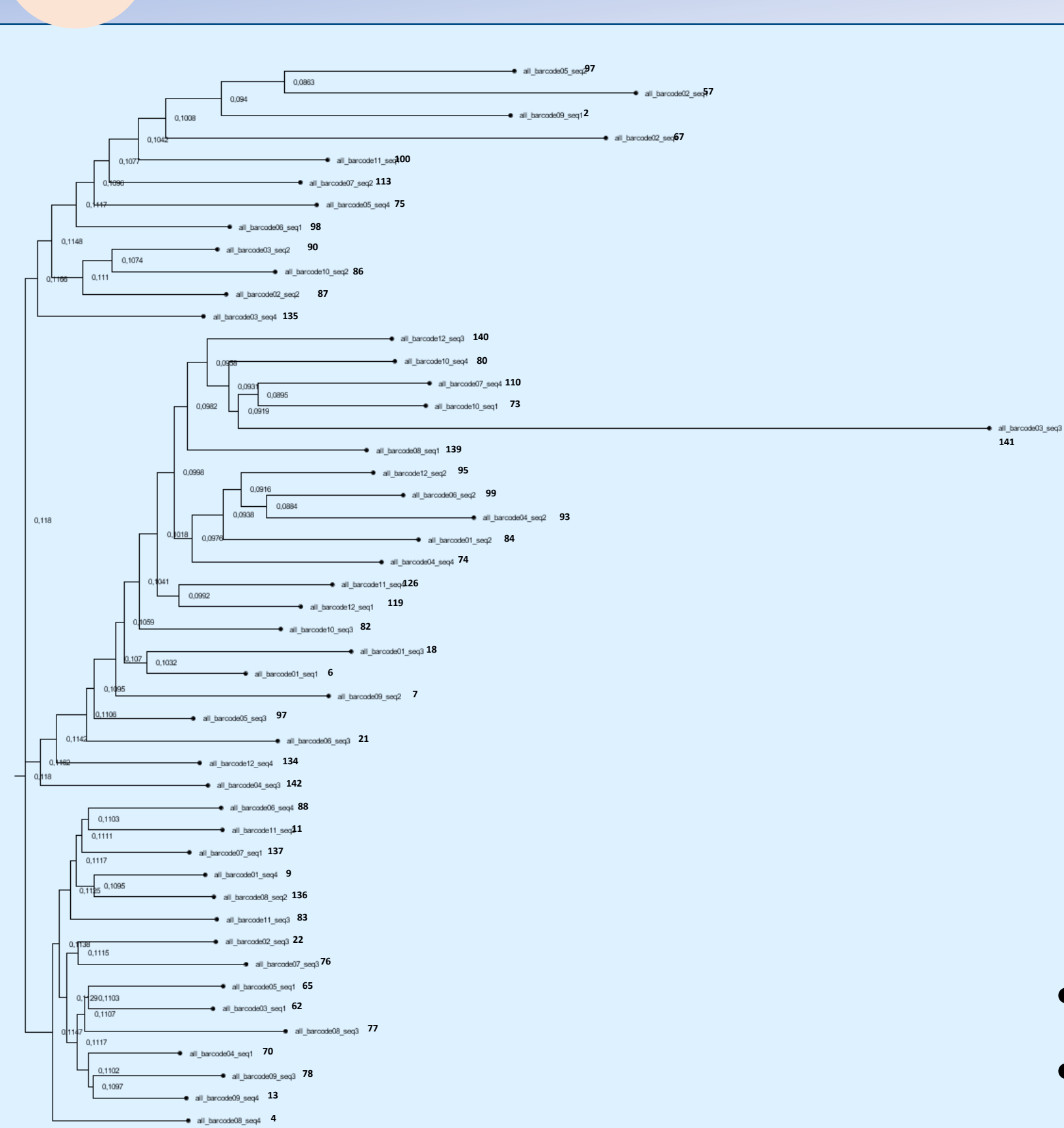
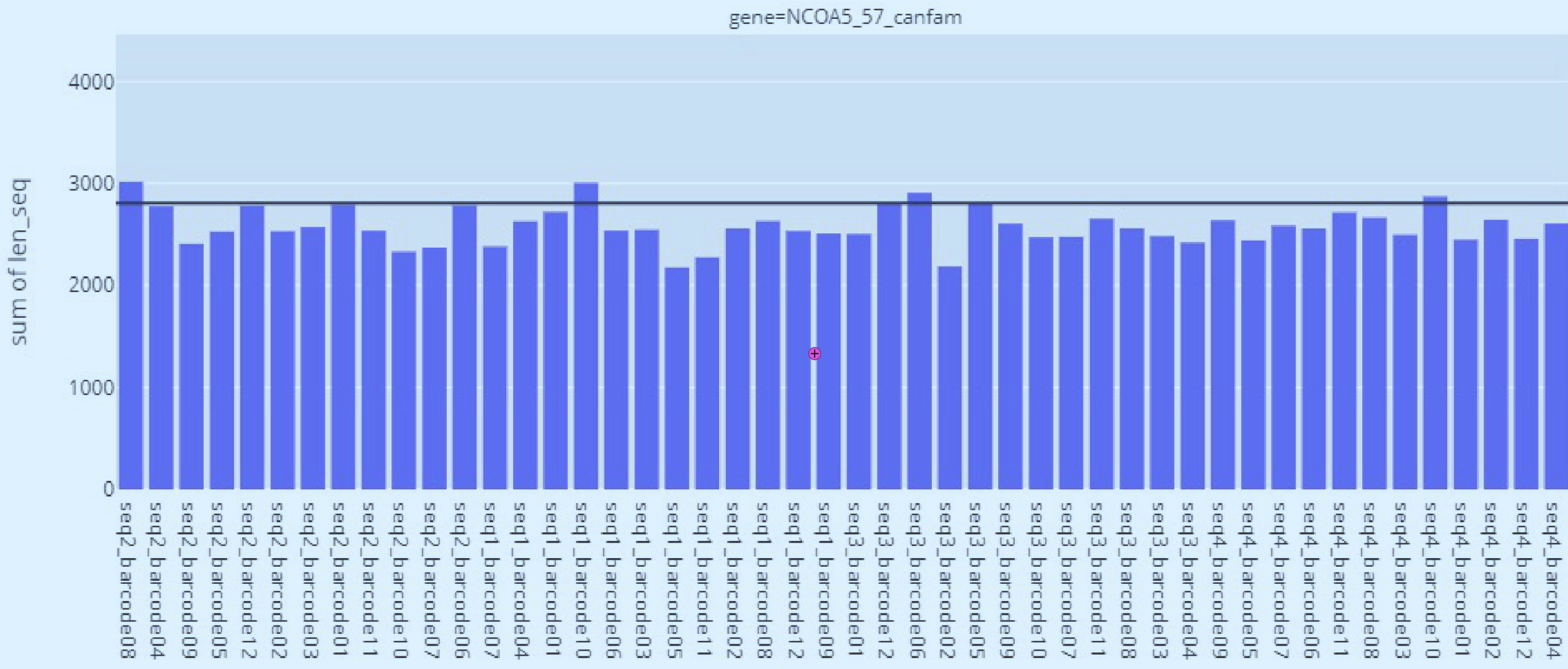


Foto: Norbert Rasch

1 Genealogy and genetic variations



Distribution of sequence lengths for the gene NCOA5 (exons 5 to 7 using the sequence from the reference genome Canfam3.1)



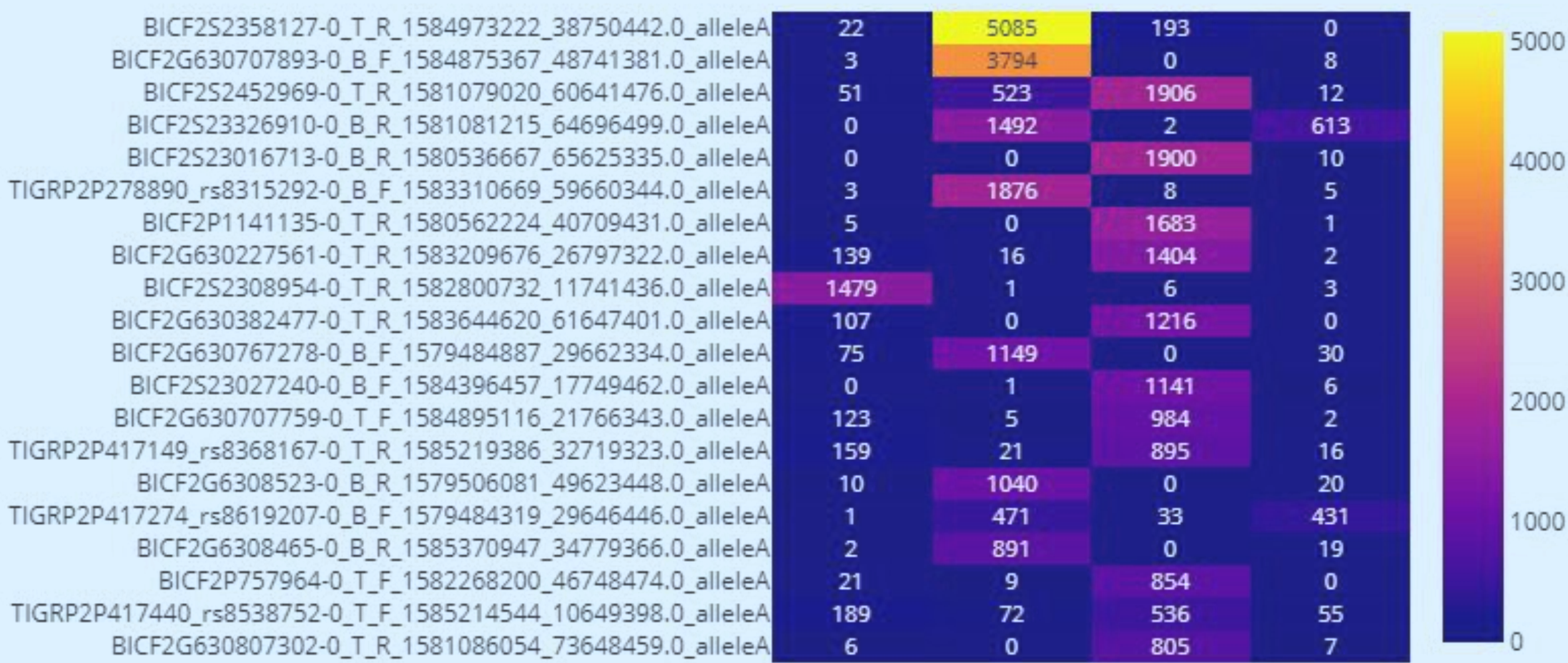
- The phylogenetic trees are not meaningful as measured by bootstrapping.
- Differences in sequence lengths, probably due to the Nanopore sequencing. Similar results for the other genes: AMHR2, RXFP2, COL2A1.

⇒ The lack of genetic information within amplicon reads of the selected gene regions (summing-up to ca. 3-kb) explain why the phylogenetic trees are not meaningful.

⇒ Need to get more information about genetic variations in the sheep poodle genome and compare it with public data from dog populations.

2 Genetic variations

Distribution of the maximum number of SNPs per probeID for all the reads



Whole dataset



Preliminary NJ clustering

⇒ As shown in the heatmap, for some SNPs, we extracted an unexpected high amount of SNP information.

⇒ This is probably due to highly overrepresented reads.

⇒ We used a subset of the obtained SNPs for comparison with SNP array results from other groups obtained with other dog breeds³.

Conclusion:

- Amplicon sequences could not be used to rebuild the pedigree provided by the breeder.
- *In silico* SNP data extraction from sequencing reads can be used to cluster dog breeds.

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References

- ¹Schön N (2021). *Genetic variants analysis of German Sheep Poodle with focus on cryptorchidism*. Master's thesis, Hochschule Mittweida
- ²Prause R (2019). *Nanopore sequencing of cryptorchidism associated genes for identification of genetic variants in German Sheep Poodle*. Master's thesis, Hochschule Mittweida
- ³Plassais J, Kim J, Davis BW et al. (2019) Whole genome sequencing of canids reveals genomic regions under selection and variants influencing morphology. Nat Commun 10:1489

Thanks to ...

